Part II: Forward Curves & Multi-Commodity Modeling

2.4 Modeling Correlation Between Commodity Markets

ESILV – Commodity Markets and Models

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Detailed Outline

1. 2.4.1 Economic Relationships: Substitutes, Complements, Inputs/Outputs
2. 2.4.2 Cointegration and Vector Error Correction Models (VECM)
3. 2.4.3 Multivariate GARCH Models: BEKK and cDCC
4. 2.4.4 Case Studies

Why Model Cross-Commodity Dependencies?

- **Real-world portfolios** involve multiple commodities
- **Risk management** requires understanding of dependencies
- **Examples**:
  - Gas-fired power plant: depends on natural gas & electricity prices
  - LNG shipping: depends on inter-continental gas price spreads
  - Oil-indexed gas contracts: depend on oil & gas prices

Important

Key Insight: Dependencies exist at two time scales:

- **Short-term**: Daily co-movement (covariance)
- **Long-term**: Equilibrium relationships (cointegration)

Different statistical techniques are needed for each!

Five Fundamental Economic Relationships

Definition

Type 1: Input-Output Relationship

- Commodity A is used to produce Commodity B
- Examples: Gas $\rightarrow$ Power, Crude Oil $\rightarrow$ Gasoline, Grains $\rightarrow$ Cattle Feed
- Dependence: Demand for A driven by price of B, supply of B driven by price of A

Example: When gas prices are high, gas-fired power plants have higher marginal costs $\rightarrow$ upward pressure on power prices

Economic Relationships (Continued)

Definition

Type 2: Substitute Goods

- Commodities A and B can be used interchangeably
- Examples: Gas vs. Coal for power generation, Gas vs. Heating Oil for heating
- Dependence: Demand for A driven by price of B (and vice versa)

Definition

Type 3: Complementary Goods

- Commodities A and B are used together to produce C
- Examples: Copper + Tin $\rightarrow$ Bronze
- Dependence: Positive correlation as price of C drives joint demand for A and B

Economic Relationships (Continued)

Definition

Type 4: Co-products

- Commodities A and B are produced together
- Examples: Gas and Oil from same field, Heavy Fuel Oil and Distillate Fuel Oil
- Dependence depends on production technology:
  - Joint production: Rise in price of B increases supply of B $\rightarrow$ increases supply of A $\rightarrow$ decreases price of A
  - Alternative production: Rise in price of B increases supply of B at expense of A $\rightarrow$ decreases supply of A $\rightarrow$ increases price of A

Economic Relationships (Continued)

Definition

Type 5: Spatial Arbitrage

- Same commodity at different locations
- Examples: Power in France vs. Germany, Gas in US vs. UK (NBP)
- Dependence linked to:
  - Volume of commodity flows between locations
  - Transportation costs between locations (Samuelson, 1952)

Pedagogical Takeaway

Remember: These relationships create dependencies at BOTH short-term and long-term horizons. Existing infrastructure creates short-term dependencies, while investment decisions create long-term dependencies.

The Bivariate Normal Distribution: A Benchmark Model

Definition

Bivariate Normal Distribution: A joint probability distribution for two random variables (X, Y) with:

$$\begin{pmatrix} X \\ Y \end{pmatrix} \sim N\left(\begin{pmatrix} \mu_X \\ \mu_Y \end{pmatrix}, \begin{pmatrix} \sigma_X^2 & \rho\sigma_X\sigma_Y \\ \rho\sigma_X\sigma_Y & \sigma_Y^2 \end{pmatrix}\right)$$

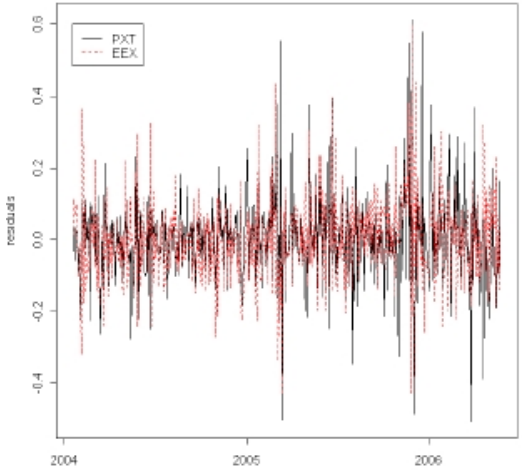
Joint density:

$$f(x, y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}} \exp\left[-\frac{z}{2(1-\rho^2)}\right]$$

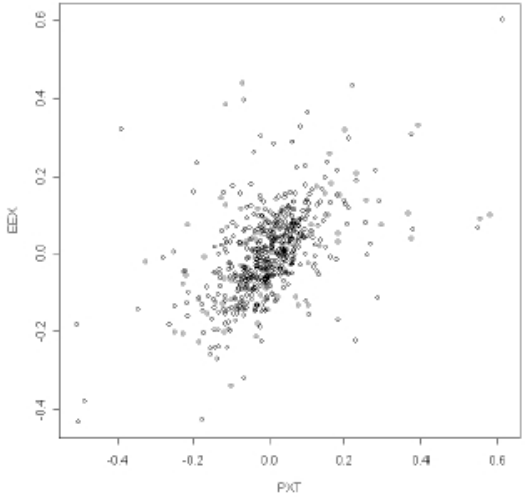
where $z = \frac{(x-\mu_X)^2}{\sigma_X^2} - \frac{2\rho(x-\mu_X)(y-\mu_Y)}{\sigma_X\sigma_Y} + \frac{(y-\mu_Y)^2}{\sigma_Y^2}$

Key Properties:

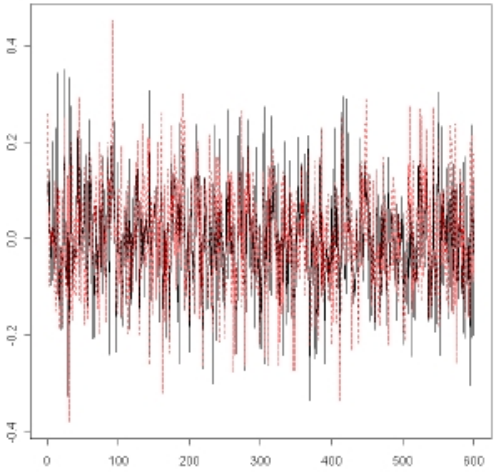
- Linear correlation ρ captures dependency
- Marginals are normal
- Elliptical contours of equal probability
- Conditional distributions are normal



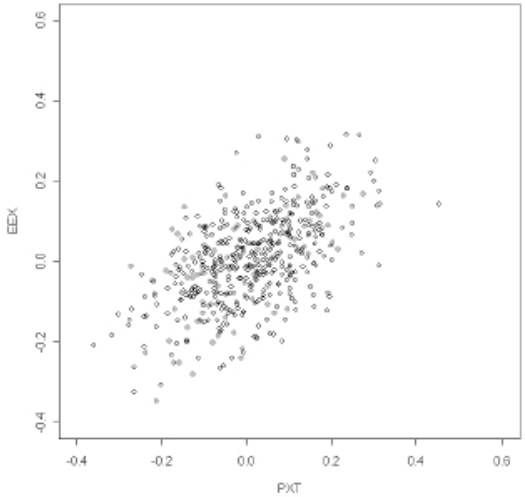
(c) Trajectory of the residuals



(d) Residuals of EEX in terms of the residuals of PXT



(e) Simulated trajectories of the residuals



(f) Simulated residuals of EEX in terms of the simulated residuals of PXT

Figure: Observed (top) vs. Simulated Normal Residuals (bottom): Normal distribution cannot capture fat tails and clustered extremes.

Mathematical Limitations of Normal Assumptions

Property	Real Returns	Bivariate Normal
Tail thickness	High (kurtosis >3)	Low (kurtosis $=3$)
Tail dependence	Often present	Zero
Volatility clustering	Present	Absent
Correlation stability	Time-varying	Constant
Skewness	Often 0	Zero
Joint extremes	Frequent	Rare

Consequences for Risk Management:

- **VaR underestimation**: Normal VaR at 99% may correspond to real 95% quantile
- **Hedge ineffectiveness**: Constant correlations miss stress-period changes
- **Option mispricing**: Underprices tail risk (out-of-the-money options)

Why Bivariate Normal Often Fails for Commodity Prices

Observed Residuals:

- Fat tails (more extremes)
- Volatility clustering
- Asymmetric dependencies

Bivariate Normal:

- Thin tails
- Constant volatility
- Symmetric dependencies

Important

Critical Limitation: Bivariate normal underestimates joint extreme events $\rightarrow$ **underestimates portfolio risk**!

Pedagogical Takeaway

Always test normality assumptions! Use: Jarque-Bera test, QQ-plots, tests for autocorrelation in squared returns.

Beyond Normality: Better Dependency Models

Three improved approaches:

1. **Copula models**: Separate marginals from dependency
   - e.g., Student-t copula with NIG margins
   - Captures tail dependence
2. **Multivariate GARCH**: Model time-varying correlations
   - e.g., BEKK, DCC models
   - Captures volatility clustering and spillovers
3. **Jump-diffusion models**: Add Poisson jumps
   - Captures extreme price movements
   - Particularly useful for electricity markets

Recommendation: For commodity pairs with frequent spikes (gas-power), start with **Student-t copula + GARCH**. For more stable relationships (crude-products), **DCC-GARCH** often suffices.

From Stationary to Non-Stationary Price Series

Definition

Stationary Series: Mean, variance, and autocovariance constant over time

- Can be modeled with VAR(p) models
- Example: Properly deseasonalized log prices

Definition

Non-Stationary (Integrated) Series: Contain stochastic trends

- VAR(p) models inappropriate
- Need **cointegration** analysis
- Example: Most commodity spot prices in levels

Important

Critical Step: Check if series is I(1) - integrated of order 1. Use:

- Augmented Dickey-Fuller (ADF) test
- Phillips-Perron (PP) test

Cointegration: The Economic Intuition

Definition

Cointegration: Two or more non-stationary series share a common stochastic trend. A linear combination of them is stationary.

Economic Interpretation: Long-run equilibrium relationship

- Crack spread: Crude oil - Gasoline prices
- Gas-oil spread: Natural gas - Crude oil prices
- Location spreads: Same commodity at different locations

Statistical Interpretation: Deviations from equilibrium are mean-reverting

Testing for Cointegration: The Johansen Procedure

Three-step approach for two series X_t^1 and X_t^2:

1. **Step 1**: Confirm both series are I(1) (unit root tests)
2. **Step 2**: Test for cointegration using Johansen test
3. **Step 3**: If cointegrated, estimate VECM

Johansen Test Statistics:

- **Trace test**: $Q_r = -T \sum_{i=r+1}^{k} \log(1 - \hat{\lambda}_i)$
- **Maximum eigenvalue test**: $\lambda_{\max} = -T \log(1 - \hat{\lambda}_{r+1})$
- Compare with critical values (non-standard distributions!)

Vector Error Correction Model (VECM)

For two cointegrated series X_t^1 and X_t^2:

$$\Delta \mathbf{Z}_t = \boldsymbol{\mu} + \sum_{k=1}^{p-1} \boldsymbol{\Gamma}_k \Delta \mathbf{Z}_{t-k} + \boldsymbol{\Pi} \mathbf{R}_{t-1} + \boldsymbol{\varepsilon}_t$$

where:

- $\mathbf{Z}_t = (X_t^1, X_t^2)'$
- $\Delta \mathbf{Z}_t = \mathbf{Z}_t - \mathbf{Z}_{t-1}$ (first differences)
- $\mathbf{R}_{t-1} = \beta_1 X_{t-1}^1 + \beta_2 X_{t-1}^2$ (cointegrating vector)
- $\boldsymbol{\Pi} = \begin{pmatrix} \pi_1 \\ \pi_2 \end{pmatrix}$ (adjustment speeds)
- $\boldsymbol{\varepsilon}_t \sim N(0, \boldsymbol{\Sigma})$

Pedagogical Takeaway

Interpretation:

- $\boldsymbol{\Pi} \mathbf{R}_{t-1}$: Error correction term - pulls system back to equilibrium
- $\boldsymbol{\Gamma}_k$: Short-term dynamics
- $\pi_i < 0$: Series i adjusts to restore equilibrium

Example: Gas-Oil Cointegration (Ohana, 2006)

Table: Linear regression: Gas log spot price on Crude oil log spot price

	Estimate	Std. Error	t-value	p-value
α	-1.92	0.030	-63.07	$< 2 \times 10^{-16}$
β	0.95	0.0089	106.04	$< 2 \times 10^{-16}$

Findings:

- $R^2 = 75.23\%$ - Strong long-term relationship
- Residuals stationary (ADF p-value $= 6\%$, PP p-value $= 3\%$)
- Cointegration confirmed!

VECM Estimation Results: Gas-Oil System

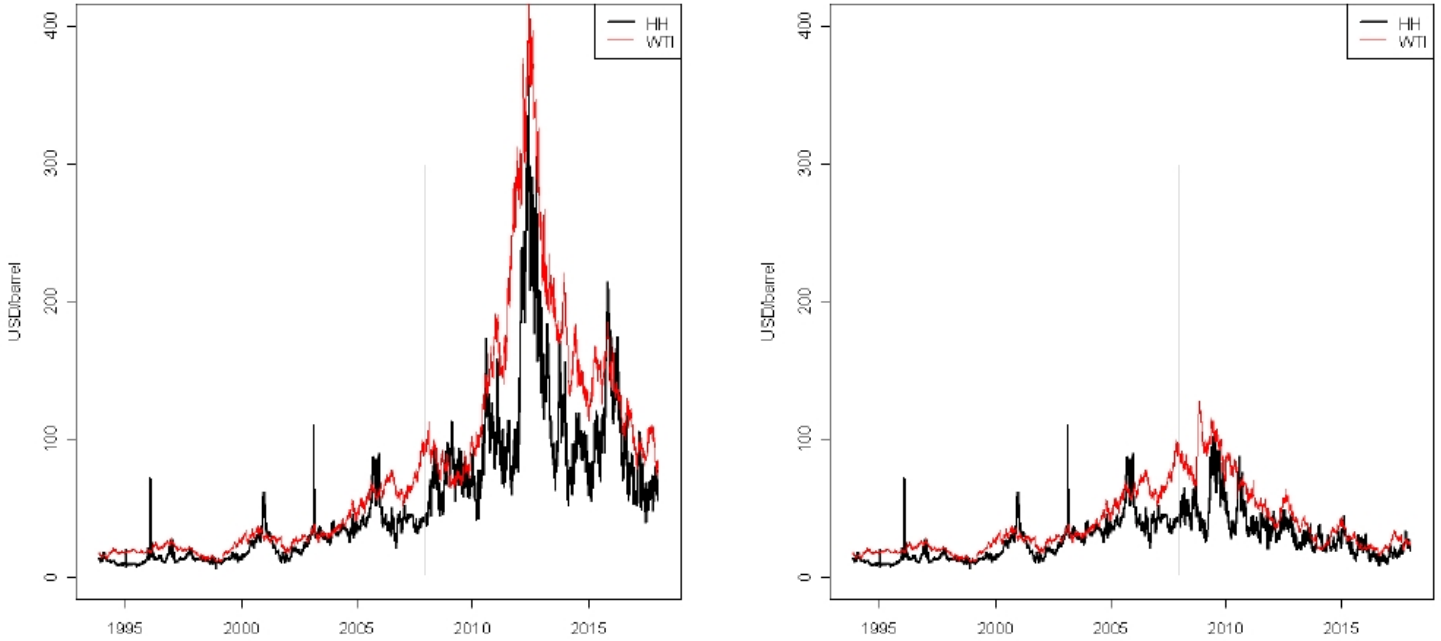
Table: VECM estimation for gas and oil returns

	Gas Equation		Oil Equation	
	Estimate	p-value	Estimate	p-value
μ	0.00029	0.77	0.00046	0.22
$\Gamma_{1,1}$	-0.045	0.006	-0.0087	0.18
$\Gamma_{1,2}$	0.15	0.00047	0.0088	0.59
Π	-0.020	3.5×10^{-9}	0.0024	0.077
R^2	1.5%		0.045%	

Pedagogical Takeaway

Crucial Findings:

- Only gas equation has significant error correction ($\Pi = -0.020$)
- Gas adjusts to oil, not vice versa $\rightarrow$ Oil is price leader
- Oil returns affect gas returns ($\Gamma_{1,2} = 0.15$ significant)
- Gas prices independent of past oil prices in long-run



(a) Realized and simulated Henry Hub and WTI spot prices under the VECM model in USD/barrel

(b) Second simulation of Henry Hub and WTI spot prices under the VECM model in USD/barrel

Figure: Two different simulated price paths under same VECM model. Shows importance of economic constraints.

Important

Risk Management Insight: VECM can generate economically implausible paths (e.g., reversion to 1990s price levels). Must filter paths using economic constraints (Hotelling model, supply-demand bounds).

Why Multivariate GARCH?

- **Volatility spillovers**: Shock in one market affects volatility in another
- **Time-varying correlations**: Dependencies change during crises
- **Risk management**: Accurate covariance matrices for portfolio VaR
- **Hedging**: Dynamic hedge ratios

Definition

Multivariate GARCH Problem: Model $\mathbf{\Sigma}_t = \mathrm{Cov}(\mathbf{X}_t | \mathcal{F}_{t-1})$ where:

$$\mathbf{X}_t = \mathbf{\Sigma}_t^{1/2} \mathbf{Z}_t, \quad \mathbf{Z}_t \sim \mathrm{WN}(0, \mathbf{I})$$

Challenge: Ensure $\mathbf{\Sigma}_t$ positive definite with reasonable parameters.

The General VEC-GARCH Model

Definition

VEC-GARCH (Bollerslev et al., 1988):

$$\text{vech}(\boldsymbol{\Sigma}_t) = \boldsymbol{\omega} + \boldsymbol{\Psi}\text{vech}(\boldsymbol{\Sigma}_{t-1}) + \boldsymbol{\Lambda}\text{vech}(\mathbf{X}_{t-1}\mathbf{X}'_{t-1})$$

where vech() stacks lower triangular elements.

Problem: Number of parameters explodes!

$$\text{Parameters} = 2\left[\frac{k(k+1)}{2}\right]^2 + \frac{k(k+1)}{2}$$

For $k = 3$: 78 parameters! Too many to estimate reliably.

Diagonal VEC-GARCH: A Simplification

Definition

Diagonal VEC-GARCH: Assume $\mathbf{\Psi}$ and $\mathbf{\Lambda}$ are diagonal:

$$\sigma_{ij,t} = \omega_{ij} + \beta_{ij}\sigma_{ij,t-1} + \alpha_{ij}X_{i,t-1}X_{j,t-1}$$

Each covariance element follows its own GARCH(1,1).

Advantages:

- $3k(k+1)/2$ parameters (for $k = 3$: 18 parameters)
- Guarantees positive definiteness under conditions

Disadvantages:

- No volatility spillovers between different elements
- Restricted dynamics

The BEKK Model: Ensuring Positive Definiteness

Definition

BEKK(p,q) (Engle & Kroner, 1995):

$$\boldsymbol{\Sigma}_t = \mathbf{C}\mathbf{C}' + \sum_{i=1}^{q} \mathbf{A}_i' \mathbf{X}_{t-i} \mathbf{X}_{t-i}' \mathbf{A}_i + \sum_{i=1}^{p} \mathbf{B}_i' \boldsymbol{\Sigma}_{t-i} \mathbf{B}_i$$

where $\mathbf{C}$ is lower triangular, $\mathbf{A}_i, \mathbf{B}_i$ are $k \times k$ matrices.

Key Properties:

- Guarantees $\boldsymbol{\Sigma}_t > 0$ (positive definite) by construction
- Allows rich volatility spillovers
- Parameters: $k^2(p+q) + k(k+1)/2$ (for BEKK(1,1) with $k = 3$: 24 parameters)

BEKK Model: Volatility Transmission

BEKK(1,1) for two assets:

$$\Sigma_t = \mathbf{C}\mathbf{C}' + \mathbf{A}'\mathbf{X}_{t-1}\mathbf{X}'_{t-1}\mathbf{A} + \mathbf{B}'\Sigma_{t-1}\mathbf{B}$$

Expanded form showing spillovers:

$$\begin{aligned}
\sigma_{11,t} &= c_{11}^2 + a_{11}^2 X_{1,t-1}^2 + 2a_{11}a_{21}X_{1,t-1}X_{2,t-1} + a_{21}^2 X_{2,t-1}^2 + \dots \\
\sigma_{12,t} &= c_{11}c_{21} + a_{11}a_{12}X_{1,t-1}^2 + (a_{11}a_{22} + a_{21}a_{12})X_{1,t-1}X_{2,t-1} + \dots \\
\sigma_{22,t} &= c_{21}^2 + c_{22}^2 + a_{12}^2 X_{1,t-1}^2 + 2a_{12}a_{22}X_{1,t-1}X_{2,t-1} + a_{22}^2 X_{2,t-1}^2 + \dots
\end{aligned}$$

Pedagogical Takeaway

Interpretation: Off-diagonal elements of **A** and **B** capture volatility spillovers. If $a_{21} \neq 0$, shocks to asset 2 affect volatility of asset 1.

Constant Conditional Correlation (CCC) Model

Definition

CCC-GARCH (Bollerslev, 1990):

$$\mathbf{\Sigma}_t = \mathbf{D}_t \mathbf{R} \mathbf{D}_t$$

where:

- $\mathbf{D}_t = \mathrm{diag}(\sigma_{1,t}, \ldots, \sigma_{k,t})$
- $\sigma_{i,t}^2 = \omega_i + \alpha_i X_{i,t-1}^2 + \beta_i \sigma_{i,t-1}^2$
- $\mathbf{R}$ is constant correlation matrix

Advantages:

- Simple: $3k + k(k-1)/2$ parameters (for $k = 3$: 12 parameters)
- Easy estimation

Disadvantage: Constant correlations unrealistic for many markets

Dynamic Conditional Correlation (DCC) Model

Definition

DCC-GARCH (Engle, 2002):

$$\mathbf{\Sigma}_t = \mathbf{D}_t \mathbf{R}_t \mathbf{D}_t$$

where $\mathbf{R}_t$ evolves as:

$$\mathbf{Q}_t = (1 - \alpha - \beta)\bar{\mathbf{Q}} + \alpha \mathbf{u}_{t-1}\mathbf{u}'_{t-1} + \beta \mathbf{Q}_{t-1}$$

with $\mathbf{u}_t = \mathbf{D}_t^{-1}\mathbf{X}_t$, and $\mathbf{R}_t = \text{diag}(\mathbf{Q}_t)^{-1/2}\mathbf{Q}_t\text{diag}(\mathbf{Q}_t)^{-1/2}$.

Advantages:

- Time-varying correlations
- Parsimonious: Only 2 parameters (α, β) for correlation dynamics
- Scalable to large portfolios

cDCC: The Corrected DCC Model

Definition

cDCC (Aielli, 2013): Corrects bias in DCC estimation

$$\mathbf{Q}_t = (1 - \alpha - \beta)\bar{\mathbf{Q}} + \alpha \mathbf{u}^*_{t-1}\mathbf{u}^{*\prime}_{t-1} + \beta \mathbf{Q}_{t-1}$$

where $\mathbf{u}^*_t = \text{diag}(\mathbf{Q}_t)^{1/2}\mathbf{u}_t$.

Key Difference from Standard DCC:

- Standard DCC uses: $\mathbf{u}_{t-1}\mathbf{u}'_{t-1}$
- cDCC uses: $\mathbf{u}^*_{t-1}\mathbf{u}^{*\prime}_{t-1} = \text{diag}(\mathbf{Q}_{t-1})^{1/2}\mathbf{u}_{t-1}\mathbf{u}'_{t-1}\text{diag}(\mathbf{Q}_{t-1})^{1/2}$

Improvements over standard DCC:

- Consistent estimation (standard DCC can be inconsistent)
- Better small-sample properties
- Now the industry standard

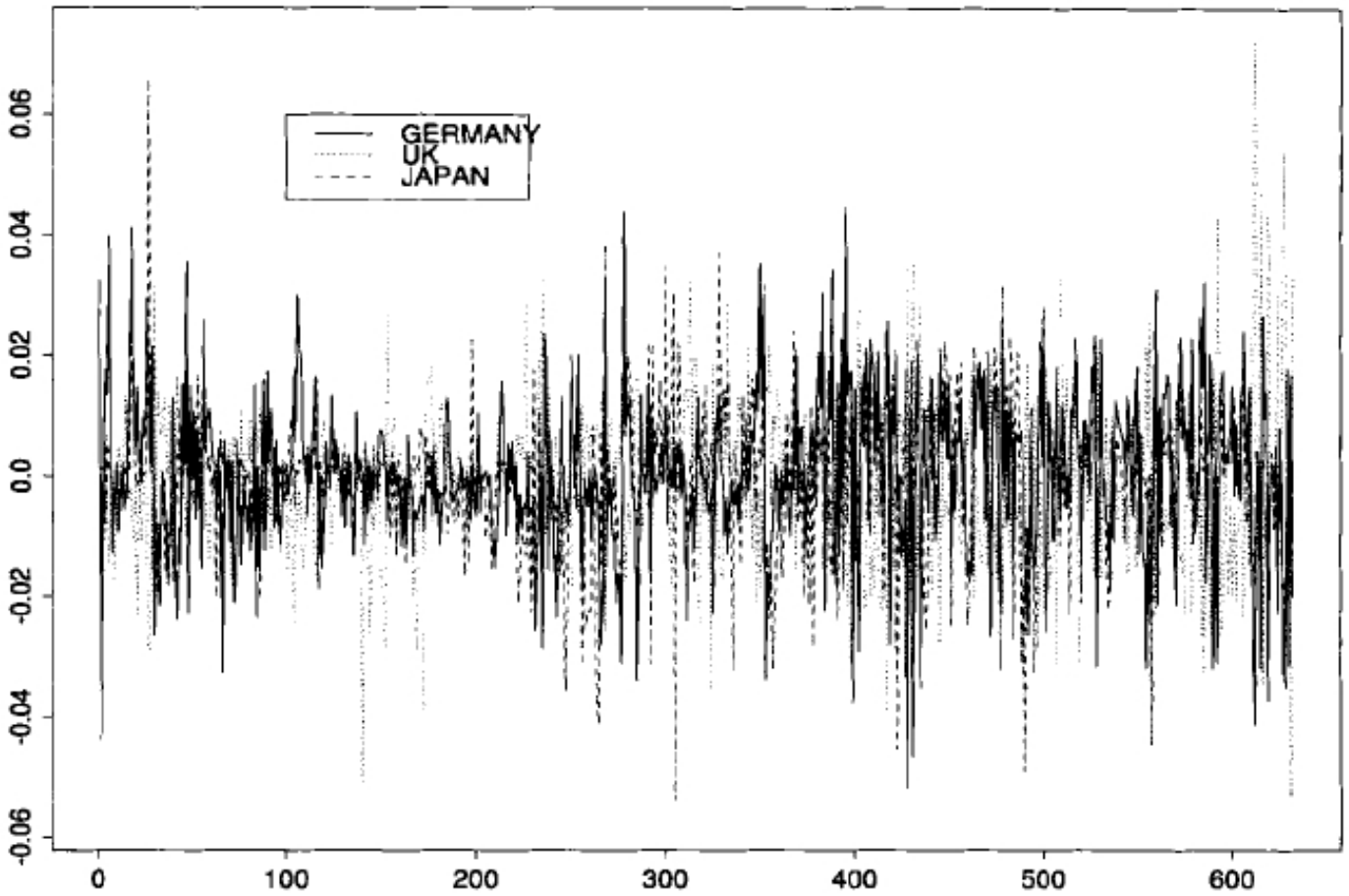


Figure: Differenced logarithms of FX series.

Data: Weekly FX rates (1973-1985), 3 currencies

- Test for stationarity (differencing needed)
- Fit multivariate GARCH models
- Compare BEKK, diagonal VEC, CCC, DCC

Table: Comparison of multivariate GARCH models

Model	# Parameters (k=3)	Strengths
VEC	78	Very general
Diagonal VEC	18	Parsimonious
BEKK	24	Positive definite, spillovers
CCC	12	Very simple
DCC/cDCC	14	Time-varying correlations, scalable

Pedagogical Takeaway

Practical Guidance:

- Small systems (2-4 assets): BEKK if spillovers important
- Medium systems (5-10 assets): cDCC for balance
- Large systems (10+ assets): cDCC or factor models

Case Study 1: Understanding the Spark Spread

Definition

Spark Spread = Profit Margin for Gas-Fired Power Plants

$$\underbrace{\text{Spark Spread}}_{\text{Profit}} = \underbrace{\text{Power Price}}_{\text{Revenue}} - \underbrace{\text{HR}}_{\text{Efficiency Factor}} \times \underbrace{\text{Gas Price}}_{\text{Cost}}$$

Heat Rate (HR) measures plant efficiency: "How much gas (MMBtu) is needed to produce 1 MWh of electricity?"

- Typical range: 7-10 MMBtu per MWh (lower = more efficient)

Simple Example: Should We Run the Plant Today?

- Power price: 40 €/MWh
- Gas price: 5 €/MMBtu
- Plant heat rate: 8 MMBtu/MWh

$$\text{Spark Spread} = 40 - (8 \times 5) = 0 \text{ €/MWh} \quad \text{(Break-even point)}$$

Interpretation: No profit margin today. Plant should stay idle.

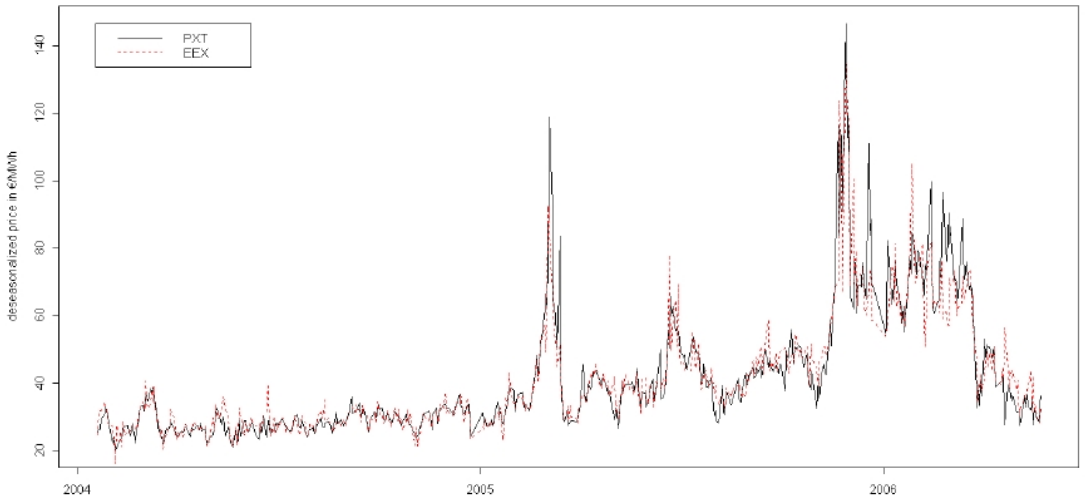
Important

Key Insight: The spark spread tells power plant operators exactly when to generate electricity:

- Positive spread: Generate (profitable)
- Negative spread: Shut down (unprofitable)

This simple calculation drives billion-euro dispatch decisions daily!

- **Market**: European power markets (PXT = France, EEX = Germany)
- **Data**: Daily deseasonalized spot prices in €/MWh
- **Objective**: Model dependency for plant valuation and optimal dispatch



Gas-Power Spark Spread: Methodology & Results

Methodology:

1. Test for stationarity of individual price series
2. Test for cointegration between gas and power prices
3. Estimate VECM for long-run equilibrium
4. Model residuals with multivariate GARCH for volatility

Key Findings (Ohana, 2006):

- Strong positive correlation (0.5) between PXT and EEX residuals
- Bi-directional Granger causality: PXT $\rightarrow$ EEX and EEX $\rightarrow$ PXT
- VAR(1) with Student-t copula captures simultaneous spikes
- **Critical**: Spread forecasting requires exogenous variables (wind, temperature, capacity margins)

Important

Practical Implication: For spark spread options valuation, must model correlated jumps and time-varying volatility!

Gas-Power Spark Spread: Multivariate GARCH Results

Volatility Spillover Analysis using BEKK model:

$$\Sigma_t = \mathbf{C}\mathbf{C}' + \mathbf{A}'\mathbf{X}_{t-1}\mathbf{X}'_{t-1}\mathbf{A} + \mathbf{B}'\Sigma_{t-1}\mathbf{B}$$

Findings:

- Significant off-diagonal elements in **A** and **B** matrices
- Volatility shocks transmit from gas to power and vice versa
- Conditional correlation: 0.3-0.7 range, time-varying
- Higher correlations during market stress periods

Pedagogical Takeaway

Plant Optimization: Dynamic hedge ratios needed! Fixed correlations underestimate risk during crises.

Case Study 2: Understanding the Crack Spread

Definition

Crack Spread = Profit Margin for Oil Refineries

$$\underbrace{\text{Crack Spread}}_{\text{Profit}} = \underbrace{\alpha \cdot \text{Gasoline} + \beta \cdot \text{Diesel}}_{\text{Revenue from Products}} - \underbrace{\text{Crude Oil}}_{\text{Cost of Input}}$$

Typical Ratios reflect refinery yields:

- **3:2:1**: 3 barrels of crude $\rightarrow$ 2 barrels of gasoline + 1 barrel of diesel
- **5:3:2**: 5 barrels of crude $\rightarrow$ 3 barrels of gasoline + 2 barrels of diesel

These ratios represent how much of each product a refinery typically produces from crude oil.

Simple Example: 3:2:1 Crack Spread

- Gasoline price: \$2.50 per gallon (42 gallons per barrel $\rightarrow$ \$105/barrel)
- Diesel price: \$2.70 per gallon $\rightarrow$ \$113.40/barrel
- Crude oil price: \$80/barrel

$$\text{Crack Spread} = (2 \times 105) + (1 \times 113.40) - (3 \times 80) = 210 + 113.40 - 240 = 83.40$$

$$\text{Per barrel margin} = 83.40/3 = 27.80 \ \$/\text{barrel}$$

- **Market**: US Gulf Coast refining
- **Data**: WTI crude, RBOB gasoline, ULSD diesel futures
- **Objective**: Risk management for refiners, crack spread option pricing

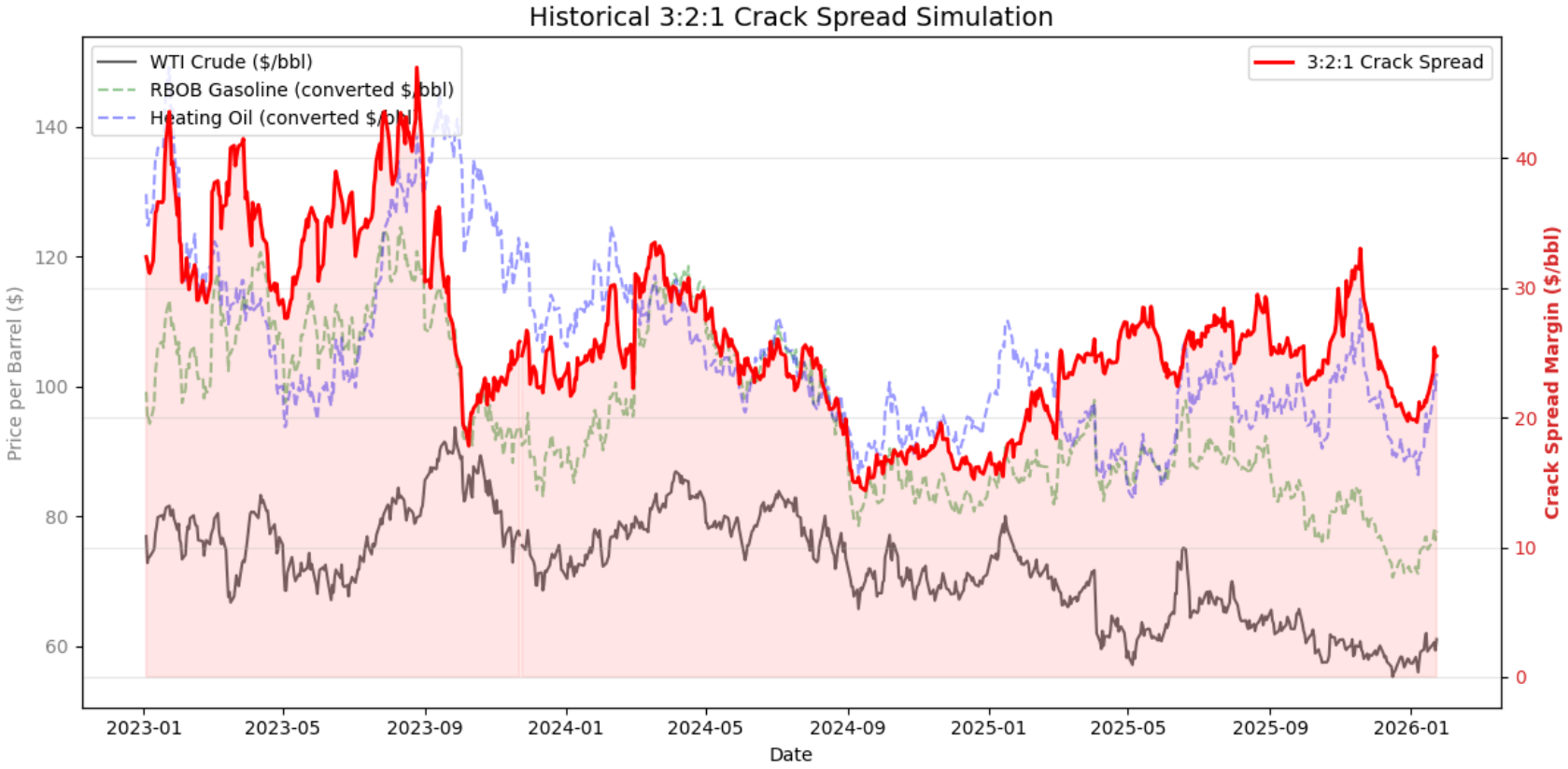


Figure: Current Estimated 3:2:1 Crack Spread: $24.76 per barrel (data from Yahoo! Finance)

Crack Spread: Cointegration Analysis

Economic Rationale: Long-run equilibrium exists due to:

- Refining technology constraints
- Storage and transportation costs (Samuelson, 1952 spatial equilibrium model)
- Substitution between gasoline and diesel

Statistical Tests:

Series	ADF Test	PP Test	I(d)
WTI Crude	-1.23	-1.19	I(1)
RBOB Gasoline	-1.45	-1.42	I(1)
ULSD Diesel	-1.37	-1.33	I(1)
3:2:1 Spread	-4.56**	-4.62**	I(0)

Conclusion: Crack spread is stationary $\rightarrow$ Cointegration exists!

Crack Spread: VECM Estimation

Three-variable VECM:

$$\Delta \mathbf{Z}_t = \boldsymbol{\mu} + \sum_{k=1}^{p-1} \boldsymbol{\Gamma}_k \Delta \mathbf{Z}_{t-k} + \boldsymbol{\Pi} \mathbf{R}_{t-1} + \boldsymbol{\varepsilon}_t$$

where $\mathbf{Z}_t = (\text{Crude}_t, \text{Gasoline}_t, \text{Diesel}_t)'$

Johansen Test Results:

- Rank $r = 2$: Two cointegrating relationships
- Cointegrating vectors:

$$\begin{aligned} \text{Vector 1:} \quad & \text{Gasoline} - 0.95 \times \text{Crude} \\ \text{Vector 2:} \quad & \text{Diesel} - 0.92 \times \text{Crude} \end{aligned}$$

- Adjustment speeds: Crude adjusts to restore both equilibria

Pedagogical Takeaway

Economic Interpretation: Crude oil is the common factor driving refined product prices in long run.

Crack Spread: MGARCH with BEKK Model

BEKK(1,1) Estimation Results:

Parameter	Crude	Gasoline	Diesel
a_{ii} (ARCH)	0.15***	0.18***	0.16***
b_{ii} (GARCH)	0.82***	0.79***	0.81***
a_{12}	0.08**		
a_{13}	0.06*		
a_{23}		0.11***	
$\rho_{12,t}$	0.65 (avg), range: 0.45-0.85		
$\rho_{13,t}$	0.60 (avg), range: 0.40-0.80		
$\rho_{23,t}$	0.75 (avg), range: 0.55-0.90		

Key Findings:

- Significant volatility spillovers (non-zero off-diagonals)
- Gasoline-diesel highest correlation (substitution effect)
- Correlations increase during supply disruptions

Crack Spread: Risk Management Applications

Value-at-Risk (VaR) for Refining Portfolio:

- **Without MGARCH**: Constant correlations underestimate tail risk
- **With MGARCH**: Captures clustering and spillovers
- **Backtest**: MGARCH VaR violations closer to expected

Option Pricing Improvement:

- Crack spread options popular hedging tool
- Black-Scholes assumes constant volatility $\rightarrow$ misprices
- MGARCH-based Monte Carlo:
  - Simulate correlated price paths
  - Include mean-reversion via VECM
  - Capture volatility clustering
- Result: More accurate option premiums

Important

Regulatory Capital: Basel III requires modeling correlation changes during stress. MGARCH essential for compliance!

Case Studies: Comparative Summary

Feature	Spark Spread	Crack Spread
Assets	Gas & Power	Crude, Gasoline, Diesel
Economic Link	Production function	Refining margin
Cointegration	Yes (1 vector)	Yes (2 vectors)
Price Leader	Gas $\rightarrow$ Power	Crude $\rightarrow$ Products
Volatility Spillovers	Strong bidirectional	Strong, especially products
Correlation Range	0.3-0.7	0.4-0.9
Typical MGARCH	BEKK or cDCC	BEKK for 3 assets
Main Application	Plant dispatch optimization	Refinery hedging

Pedagogical Takeaway

- **Common Pattern**: Cointegration + MGARCH works for both spreads
- **Difference**: Crack spread has more complex cointegration (3 assets, 2 vectors)
- **Unified Approach**: Test for economic relationship $\rightarrow$ Cointegration $\rightarrow$ VECM $\rightarrow$ MGARCH residuals

Summary: Key Takeaways

1. **Economic relationships** determine statistical dependencies
2. **Cointegration** captures long-run equilibria; use VECM for error correction
3. **MGARCH** models time-varying volatility and correlations
4. **BEKK** ensures positive definiteness and allows spillovers
5. **cDCC** offers scalable, time-varying correlations
6. Always: **Test assumptions**, **Check diagnostics**, **Consider economic plausibility**

Important

Final Warning: No model is perfect. The "best" model depends on:

- Your specific commodities
- Available data
- Purpose (risk management, trading, valuation)
- Computational constraints

References & Further Reading

- **Ohana (2006)**: "Modeling correlation between commodity markets" - Source of economic relationships and cointegration examples
- **Chan (2010)**: *Time Series: Applications to Finance with R and S-Plus* - Source of MGARCH examples and implementation
- **Engle & Kroner (1995)**: "Multivariate simultaneous generalized ARCH" - BEKK model
- **Engle (2002)**: "Dynamic conditional correlation" - DCC model
- **Aielli (2013)**: "Dynamic conditional correlation with corrected correlations" - cDCC model
- **Johansen (1995)**: *Likelihood-Based Inference in Cointegrated VAR Models* - Cointegration testing
- **Bauwens, Laurent & Rombouts (2006)**: "Multivariate GARCH models: a survey" - Comprehensive MGARCH review